PDF-09 — KPIs, Analytics & Reporting

Status: Locked (MVP)
Scope: SAT only
Audience: Engineering, Product, Data, QA, Growth
Depends on:

- PDF-02 (Canonical Questions & IDs)
- PDF-03 (Practice Engine)
- PDF-04 (Full-Length Exams & Scoring)
- PDF-05 (Adaptive & Mastery Engine)
- PDF-06 (AI Tutor)
- PDF-07 (Calendar & Study Plan)
- PDF-08 (Auth, Roles & Entitlements)

Purpose:
Define a KPI system that is **honest, explainable, non-gameable, and useful** for students, guardians, and the product itself—without vanity metrics or misleading claims.

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1. KPI Philosophy (Non-Negotiable)

1. **No vanity metrics** (e.g., “AI confidence score”).
2. **Every metric must be explainable to a parent.**
3. **Metrics must align with student experience.**
4. **Raw ≠ Weighted ≠ Official. These are never conflated.**
5. **Metrics inform decisions, not punish behavior.**

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2. KPI Audiences

KPIs are strictly segmented by role.

2.1 Student KPIs

- Learning progress
- Readiness signals
- Actionable insights

2.2 Guardian KPIs

- High-level progress
- Consistency and effort
- Outcome-oriented summaries

2.3 Internal KPIs (Lyceon only)

- System health
- Learning efficacy
- Cost and engagement efficiency

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3. Student Dashboard KPIs (Authoritative)

3.1 Mastery Overview

- **Domain Hexagon** (from PDF-05)
- Labels only (Not Started → Strong)
- Directional change indicators (↑ ↓ →)

Purpose:
Instant understanding of strengths and gaps.

3.2 Practice Performance

Metric	Definition
Accuracy	% correct (rolling window)
Questions Attempted	Count
Avg Time / Question	Latency
Review Recovery Rate	% of missed questions fixed in review

Notes

- Accuracy is contextual (practice only).
- Review recovery is highlighted as *learning*, not failure.

3.3 Test Readiness (Student-Facing)

Metric	Meaning
Last Scaled Score	From full test
Score Trend	Up / Flat / Down
Weak Domains	Derived from test + mastery
Confidence Band	Low / Medium / High (derived, not predicted)

No “predicted score” language in MVP.

3.4 Consistency Signals

Metric	Description

Days Studied / Week	Calendar adherence
Sessions Completed	Count
Skipped Blocks	Informational only

No streak shaming. No penalties.

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4. Guardian Dashboard KPIs (Authoritative)

Guardians see **summaries, not internals**.

4.1 Outcome Summary

- Latest full test score
- Change since last test
- Section-level summary (M vs RW)

4.2 Learning Progress

Metric	Meaning
Domains Improving	Count
Domains Needing Focus	Count
Review Follow-Through	High / Medium / Low

Language is deliberately non-judgmental.

4.3 Engagement & Consistency

Metric	Meaning

Weekly Study Time	Approximate
Plan Adherence	% of blocks completed
Missed Weeks	Informational

Guardians never see:

- question-level data
- mastery scores
- tutor interactions
- raw deltas

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5. Full Test Analytics (Student + Guardian)

5.1 Raw Score Breakdown

- Correct / total by section
- Correct / total by domain
- Correct / total by skill (student only)

5.2 Scaled Score

- Official SAT-modeled scaling
- Clear disclosure that real SAT forms vary

5.3 Diagnostic Signals (Clearly Labeled)

- “High-impact weaknesses”
- “Likely score blockers”

These are **learning diagnostics**, not guarantees.

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6. KPI Computation Rules (Critical)

6.1 Rolling Windows

- Practice metrics use recent N attempts
- Review metrics use last M reviews
- Test metrics are per test (no averaging)

6.2 No Hidden Weighting

- Any weighting applied (difficulty, recency, test context) must:
    - be documented
    - be internally consistent
    - never be hidden from explanations

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7. Analytics Data Model (Conceptual)

```
kpi_snapshots (
  id uuid primary key,
  student_id uuid,
  snapshot_type text,          -- student | guardian | internal
  data jsonb,
  generated_at timestamptz
)
```

Snapshots are regenerated, not mutated.

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8. Explainability Layer (Required)

Every KPI displayed must have:

- "What this means"
- "Why this changed"
- "What to do next"

Example:

> "Your Algebra progress improved because you corrected most mistakes during review this week."

This text is rule-based, not LLM-invented.

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9. KPI → Action Mapping

KPIs must map to **clear next steps**:

Signal	Suggested Action
Low review recovery	More Review blocks
Weak test domain	Targeted practice
High effort, flat score	Strategy + tutor

This prevents dashboards from becoming passive charts.

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10. Entitlement Gating

KPI Type	Free	Paid
Basic practice stats	✅	✅
Mastery hexagon	❌	✅

Full test analytics	❌	✅
Historical trends	❌	✅

Guardians inherit access only if the student is entitled.

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11. Internal KPIs (Non-User Facing)

Used to improve Lyceon, never shown externally.

- Tutor success rate
- Review recovery effectiveness
- Time-to-mastery per skill
- Cost per learning gain
- Drop-off correlation vs workload

These guide roadmap decisions.

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12. Privacy & Safety

- No PII in analytics payloads
- Aggregation before storage
- No cross-student comparisons shown
- No leaderboard or ranking features

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13. Failure Modes

Issue	Handling
Missing data	Graceful fallback
Inconsistent signals	Prefer test truth
Metric confusion	Improve explanation
Guardian overreach	Enforce role gating

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14. Acceptance Criteria

This feature is complete when:

- KPIs match student intuition
- Guardians trust summaries
- No metric feels punitive
- Explanations are clear
- Analytics feed planning and adaptation
- No vanity or misleading numbers exist

End of PDF-09

Next

Say **“Proceed to PDF-10 — SEO, Marketing, Trust & Growth Surfaces.”**

That final feature PDF will cover:

- SEO architecture
- landing pages

- trust signals for parents
- conversion surfaces
- analytics-safe growth loops

pdf 10

Below is **PDF-10 — SEO, Marketing, Trust & Growth Surfaces**, written end-to-end and aligned with PDFs 02–09.
This document defines **how Lyceon is discovered, trusted, and converted**—without compromising learning integrity, privacy, or regulatory posture.

You can paste this directly into a document and export as PDF.

PDF-10 — SEO, Marketing, Trust & Growth Surfaces

Status: Locked (MVP)
Scope: SAT only
Audience: Engineering, Product, Growth, Legal, QA
Depends on:

- PDF-02 (Canonical Questions & IDs)
- PDF-03 (Practice Engine)
- PDF-04 (Full-Length Exams & Scoring)
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- PDF-06 (AI Tutor)
- PDF-07 (Calendar & Study Plan)
- PDF-08 (Auth, Roles & Entitlements)
- PDF-09 (KPIs & Analytics)

Purpose:
Define how Lyceon attracts users, builds trust (especially with parents), converts ethically, and grows—while remaining compliant, honest, and technically sound.

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1. Growth Philosophy (Non-Negotiable)

1. **Trust before conversion**
2. **Accuracy over hype**
3. **Parents are decision-makers**
4. **Students are the users**
5. **Growth must not distort learning signals**

Lyceon does not sell “guaranteed scores.” It sells **clarity, structure, and measurable progress**.

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2. Primary Growth Surfaces

2.1 Public Website (SEO-Indexed)

- Homepage
- Product pages
- SAT learning guides
- Parent explainer pages
- Blog / resources

2.2 In-Product Surfaces

- Soft paywalls

- Progress previews
- Test readiness messaging
- Guardian dashboards

2.3 External

- Search (SEO)
- Referrals (future)
- School / counselor partnerships (future)

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3. SEO Architecture (Authoritative)

3.1 Page types

Page Type	Purpose
`/`	Product overview
`/sat`	SAT hub
`/sat/math`	Section explainer
`/sat/reading-writing`	Section explainer
`/parents`	Guardian trust page
`/blog/*`	Long-form SEO content

3.2 Technical SEO Requirements

- Server-side rendering (or pre-render)
- Canonical URLs

- Sitemap.xml (auto-generated)
- Robots.txt (explicit allow)
- Clean URL slugs
- Page speed optimized
- Accessible markup (ARIA, alt text)

3.3 Structured Data (Schema.org)

- `Product`
- `FAQPage`
- `EducationalOrganization`
- `BreadcrumbList`

No fake reviews. No star manipulation.

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4. Content Strategy (SAT-Aligned)

4.1 What we publish

- “How SAT scoring works”
- “Why practice alone isn’t enough”
- “How to interpret practice vs test scores”
- “How parents can support SAT prep”

4.2 What we never claim

- Guaranteed score increases
- Official SAT endorsement
- “AI predicts your SAT score” (MVP)

All claims are defensible and documented.

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5. Trust Surfaces (Critical for Parents)

5.1 Trust pillars

- **Transparency**
  How scoring works, what’s official vs diagnostic.
- **Explainability**
  Why the plan looks the way it does.
- **Control**
  Students can override plans; nothing is forced.
- **Privacy**
  No selling of data. No ads. No third-party trackers.

5.2 Parent-Facing Pages

- “How we calculate scores”
- “How mastery works”
- “What the AI tutor does (and doesn’t do)”
- “What data we collect”

These pages are written in plain English, linked from the product.

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6. Conversion Strategy (Ethical)

6.1 Free tier value

- Real practice
- Real feedback
- Limited analytics
- No dark patterns

6.2 Paid unlocks

- Full tests
- Mastery hexagon
- Adaptive calendar
- Advanced analytics
- Tutor depth

Conversion is framed as:

> “Unlock deeper insight,” not “fix your weaknesses.”

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7. Paywalls & Upgrade Moments

7.1 Allowed paywall triggers

- Attempting full test
- Viewing mastery breakdown
- Viewing historical trends

- Guardian viewing student data

7.2 Paywall rules

- Always explain **what unlocks**
- Never block already-earned data
- Never interrupt an active test

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8. Messaging & Copy Guardrails

8.1 Approved language

- “SAT-aligned”
- “Official scoring methodology (modeled)”
- “Learning diagnostics”
- “Practice-based insights”

8.2 Prohibited language

- “Guaranteed”
- “Official SAT”
- “Predicted score” (MVP)
- “Perfect score”

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9. Analytics for Growth (Privacy-Safe)

9.1 Allowed tracking

- Page views
- Funnel steps
- Feature usage
- Conversion events

9.2 Disallowed

- Session replay with content
- Keystroke logging
- Cross-site tracking
- Selling data

All analytics are **aggregated and anonymized**.

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10. Referral & Virality (Future-Safe)

10.1 Referral logic (future)

- Student invites guardian
- Guardian invites student
- Credits unlock features (not answers)

10.2 Safeguards

- No pressure sharing
- No public leaderboards
- No “invite spam”

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11. Legal & Compliance (Surface-Level)

11.1 Required pages

- Terms of Service
- Privacy Policy
- Guardian Consent Notice
- Refund Policy

11.2 Claims review

All marketing copy must pass:

- Legal review
- Product truth check
- Parent-readability check

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12. Failure Modes

Risk	Mitigation
Over-promising	Copy guardrails
SEO misinformation	Source-linked content
Parent distrust	Transparency pages
Dark patterns	Explicit prohibition
Data misuse	Minimal collection

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13. Acceptance Criteria

This feature is complete when:

- SEO pages are crawlable and fast
- Messaging matches product reality
- Parents understand what Lyceon does
- Conversion is ethical and explainable
- Growth does not distort learning
- Legal posture is defensible